PROBLEM STATEMENT

DASHBOARD 1: SUMMARY

Key Performance Indicators (KPIs) Requirements:

1. **Total Loan Applications:** We need to calculate the total number of loan applications received during a specified period. Additionally, it is essential to monitor the Month-to-Date (MTD) Loan Applications and track changes Month-over-Month (MoM).
2. **Total Funded Amount:** Understanding the total amount of funds disbursed as loans is crucial. We also want to keep an eye on the MTD Total Funded Amount and analyse the Month-over-Month (MoM) changes in this metric.
3. **Total Amount Received:** Tracking the total amount received from borrowers is essential for assessing the bank's cash flow and loan repayment. We should analyse the Month-to-Date (MTD) Total Amount Received and observe the Month-over-Month (MoM) changes.
4. **Average Interest Rate:** Calculating the average interest rate across all loans, MTD, and monitoring the Month-over-Month (MoM) variations in interest rates will provide insights into our lending portfolio's overall cost.
5. **Average Debt-to-Income Ratio (DTI):** Evaluating the average DTI for our borrowers helps us gauge their financial health. We need to compute the average DTI for all loans, MTD, and track Month-over-Month (MoM) fluctuations.

PROBLEM STATEMENT

DASHBOARD 1: SUMMARY

Good Loan v Bad Loan KPI's

Good Loan:

1. Good Loan Application Percentage
2. Good Loan Applications
3. Good Loan Funded Amount
4. Good Loan Total Received Amount

Bad Loan

1. Bad Loan Application Percentage
2. Bad Loan Applications
3. Bad Loan Funded Amount
4. Bad Loan Total Received Amount

Loan Status Grid View

In order to gain a comprehensive overview of our lending operations and monitor the performance of loans, we aim to create a grid view report categorized by 'Loan Status.' By providing insights into metrics such as 'Total Loan Applications,' 'Total Funded Amount,' 'Total Amount Received,' 'Month-to-Date (MTD) Funded Amount,' 'MTD Amount Received,' 'Average Interest Rate,' and 'Average Debt-to-Income Ratio (DTI),' this grid view will empower us to make data-driven decisions and assess the health of our loan portfolio.

PROBLEM STATEMENT

DASHBOARD 2: OVERVIEW

CHARTS

1. **Monthly Trends by Issue Date (Line Chart):** To identify seasonality and long-term trends in lending activities
2. **Regional Analysis by State (Filled Map):** To identify regions with significant lending activity and assess regional disparities
3. **Loan Term Analysis (Donut Chart):** To allow the client to understand the distribution of loans across various term lengths.
4. **Employee Length Analysis (Bar Chart):** How lending metrics are distributed among borrowers with different employment lengths, helping us assess the impact of employment history on loan applications.
5. **Loan Purpose Breakdown (Bar Chart):** **W**ill provide a visual breakdown of loan metrics based on the stated purposes of loans, aiding in the understanding of the primary reasons borrowers seek financing.
6. **Home Ownership Analysis (Tree Map):** For a hierarchical view of how home ownership impacts loan applications and disbursements.

Metrics to be shown: 'Total Loan Applications,' 'Total Funded Amount,' and 'Total Amount Received'

PROBLEM STATEMENT

DASHBOARD 3: DETAILS

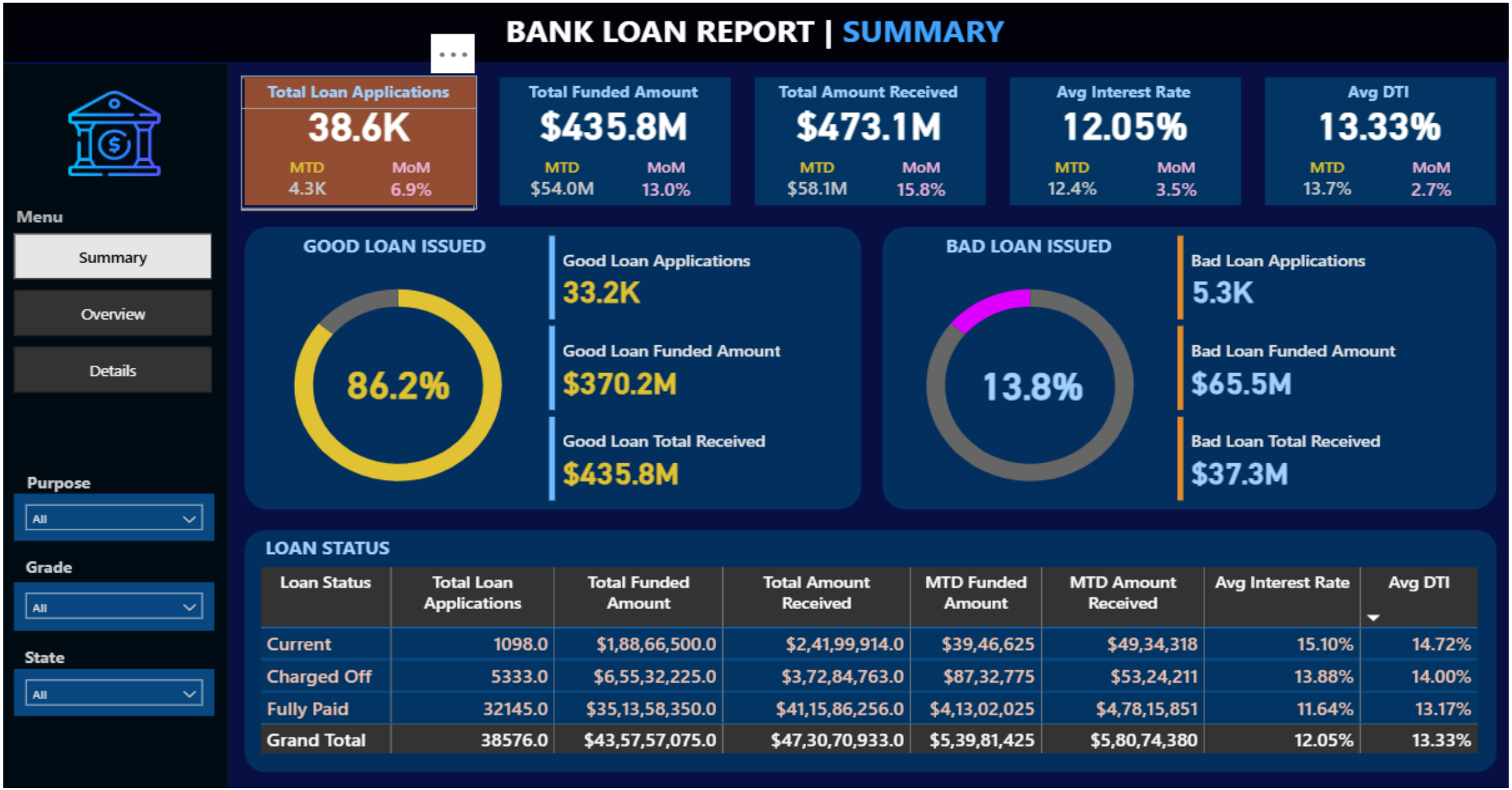
GRID

Need for a comprehensive 'Details Dashboard' that provides a consolidated view of all the essential information within our loan data. This Details Dashboard aims to offer a holistic snapshot of key loan-related metrics and data points, enabling users to access critical information efficiently.

Objective:

The primary objective of the Details Dashboard is to provide a comprehensive and user-friendly interface for accessing vital loan data. It will serve as a one-stop solution for users seeking detailed insights into our loan portfolio, borrower profiles, and loan performance.

POWER BI



Loan Status	Total Loan Applications	Total Funded Amount	Total Amount Received	MTD Funded Amount	MTD Amount Received	Avg Interest Rate	Avg DTI
Current	1098.0	$1,88,66,500.0	$2,41,99,914.0	$39,46,625	$49,34,318	15.10%	14.72%
Charged Off	5333.0	$6,55,32,225.0	$3,72,84,763.0	$87,32,775	$53,24,211	13.88%	14.00%
Fully Paid	32145.0	$35,13,58,350.0	$41,15,86,256.0	$4,13,02,025	$4,78,15,851	11.64%	13.17%
Grand Total	38576.0	$43,57,57,075.0	$47,30,70,933.0	$5,39,81,425	$5,80,74,380	12.05%	13.33%

POWER BI

BANK LOAN REPORT | OVERVIEW
Menu
Summary
Overview
Details
Select Measure
Total Funded Amount
Good v Bad Loan
All
Grade
All
State
All
Total Loan Applications
38.6K
MTD 4.3K
MoM 6.9%
Total Funded Amount
$435.8M
MTD $54.0M
MoM 13.0%
Total Amount Received
$473.1M
MTD $58.1M
MoM 15.8%
Avg Interest Rate
12.05%
MTD 12.4%
MoM 3.5%
Avg DTI
13.33%
MTD 13.7%
MoM 2.7%
Total Funded Amount by Month
$25M $25M $29M $30M $32M $34M $36M $38M $41M $45M $48M $54M
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Total Funded Amount by State
Total Funded Amount by Term
36 months 60 months
$162.7M (37.34%)
$273.0M (62.66%)
Total Funded Amount by Employee Length
10+ years $116M
2 years $45M
< 1 year $44M
3 years $44M
4 years $38M
5 years $37M
1 year $33M
6 years $26M
Total Funded Amount by Purpose
Debt consoli... $0.23bn
credit card $0.06bn
home impro... $0.03bn
other $0.03bn
small busine... $0.02bn
major purch... $0.02bn
car $0.01bn
wedding $0.01bn
Total Funded Amount by home_ownership
MORTGAGE
$219.3M
RENT
$185.8M

POWER BI

BANK LOAN REPORT | DETAILS

Menu

- Summary
- Overview
- Details

Good v Bad Loan: All

Grade: All

State: All

Total Loan Applications	MTD	MoM
38.6K	4.3K	6.9%

Total Funded Amount	MTD	MoM
$435.8M	$54.0M	13.0%

Total Amount Received	MTD	MoM
$473.1M	$58.1M	15.8%

Avg Interest Rate	MTD	MoM
12.05%	12.4%	3.5%

Avg DTI	MTD	MoM
13.33%	13.7%	2.7%

id	purpose	home_ownership	grade	sub_grade	issue_date	Funded Amount	Int Rate	Sum of installment	Amount Collection
54734	Debt consolidation	RENT	B	B4	09 August 2021	$25,000.0	11.89%	$829.1	$29,330.0
55742	credit card	RENT	B	B5	08 May 2021	$7,000.0	10.71%	$228.2	$8,216.0
57245	Debt consolidation	OWN	C	C2	10 March 2021	$1,200.0	13.11%	$40.5	$1,458.0
57416	Debt consolidation	RENT	C	C3	09 November 2021	$10,800.0	13.57%	$366.9	$13,208.0
58915	Debt consolidation	RENT	B	B3	08 April 2021	$7,500.0	10.08%	$162.3	$5,844.0
59006	credit card	MORTGAGE	C	C5	09 September 2021	$3,000.0	14.26%	$102.9	$3,705.0
61390	credit card	MORTGAGE	A	A5	10 February 2021	$4,000.0	7.88%	$125.1	$4,452.0
61419	Debt consolidation	RENT	D	D2	10 February 2021	$5,600.0	14.96%	$194.0	$6,475.0
62102	Debt consolidation	RENT	B	B1	10 April 2021	$3,200.0	9.88%	$103.1	$3,414.0
65426	car	MORTGAGE	B	B1	09 August 2021	$4,000.0	11.14%	$131.2	$2,755.0
65640	home improvement	MORTGAGE	C	C2	08 May 2021	$5,000.0	11.34%	$87.2	$3,154.0
66431	Debt consolidation	RENT	B	B5	09 February 2021	$2,525.0	12.21%	$84.1	$3,028.0
66749	Debt consolidation	MORTGAGE	C	C4	08 December 2021	$10,625.0	13.47%	$360.4	$12,975.0
66943	Debt consolidation	RENT	B	B4	10 August 2021	$2,800.0	11.49%	$61.6	$3,144.0
66964	Debt consolidation	MORTGAGE	D	D3	08 June 2021	$7,500.0	13.24%	$253.6	$9,129.0
67503	Debt consolidation	MORTGAGE	A	A4	09 October 2021	$10,000.0	8.59%	$316.1	$11,280.0
68163	small business	MORTGAGE	A	A3	10 February 2021	$3,000.0	7.14%	$92.8	$3,342.0
68381	Debt consolidation	RENT	A	A5	08 March 2021	$6,625.0	8.63%	$209.5	$7,542.0